

3614. Process String with Special Operations II

Difficulty: Hard

Link: <https://leetcode.com/problems/process-string-with-special-operations-ii/?envType=daily-question&envId=2026-06-17>

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Problem:

You are given a string `s` consisting of lowercase English letters and the special characters: `'*'`, `'#'`, and `'%'`.

You are also given an integer `k`.

Build a new string `result` by processing `s` according to the following rules from left to right:

- If the letter is a **lowercase** English letter append it to `result`.
- A `'*'` **removes** the last character from `result`, if it exists.
- A `'#'` **duplicates** the current `result` and **appends** it to itself.
- A `'%'` **reverses** the current `result`.

Return the k^{th} character of the final string `result`. If `k` is out of the bounds of `result`, return `'.'`.

Example 1:

Input: `s = "a#b%*", k = 1`

Output: `"a"`

Explanation:

i	s[i]	Operation	Current result
0	'a'	Append 'a'	"a"
1	'#'	Duplicate result	"aa"
2	'b'	Append 'b'	"aab"
3	'%'	Reverse result	"baa"
4	'*'	Remove the last character	"ba"

The final result is "ba" . The character at index k = 1 is 'a' .

Example 2:

Input: s = "cd%#*#", k = 3

Output: "d"

Explanation:

i	s[i]	Operation	Current result
0	'c'	Append 'c'	"c"
1	'd'	Append 'd'	"cd"

i	s[i]	Operation	Current result
2	'%'	Reverse result	"dc"
3	'#'	Duplicate result	"dcdc"
4	'*'	Remove the last character	"dcd"
5	'#'	Duplicate result	"dcdcdc"

The final result is "dcdcdc". The character at index $k = 3$ is 'd'.

Example 3:

Input: $s = "z*#", k = 0$

Output: "."

Explanation:

i	s[i]	Operation	Current result
0	'z'	Append 'z'	"z"
1	'*'	Remove the last character	"
2	'#'	Duplicate the string	"

The final result is "". Since index $k = 0$ is out of bounds, the output is '.'.

Constraints:

- $1 \leq s.length \leq 10^5$
- `s` consists of only lowercase English letters and special characters `'*'`, `'#'`, and `'%'`.
- $0 \leq k \leq 10^{15}$
- The length of `result` after processing `s` will not exceed 10^{15} .